



Swarm Digital Twin (Gazebo 8 on ASUS TUF) LIDAR & Mesh Active



3D GIS Dashboard (React + Mapbox) WebGPU HUD Stream

01 Option A: Single Monorepo + Dedicated Member Folders & Branches

<https://github.com/Team-Offgrid/SUTRA.git> (Single Unified Monorepo)

sutra_ws/src/sutra_gnc
Rohith (feature/gnc)

sutra_ws/src/sutra_comms
Nikhil (feature/comms)

sutra_ws/src/sutra_perception
Vedanth (feature/vision)

sutra_ws/src/sutra_gcs
Siva Kesava (feature/gcs)

docs/ & sutra_sim/
Harika (feature/docs)

02 Dedicated Role Ownership & Code Folder Assignments

ROHITH KUMAR (Subsystem A: GNC)

sutra_gnc/

Role Analogy: "The Steering Wheel & Brakes".
Builds: px4_offboard_node.cpp (PX4 Offboard flight trajectory & ORCA collision avoidance).
Git Branch: feature/subsystem-a-gnc

NIKHIL (Subsystem B: Comms & Tech Lead)

sutra_comms/

Role Analogy: "The Walkie-Talkie & Brain".
Builds: mesh_socket.py (802.11s mesh socket, Deep JSCC neural encoder & Gazebo Sim ops).
Git Branch: feature/subsystem-b-comms

VEDANTH SAI RAM (Subsystem C: Vision)

sutra_perception/

Role Analogy: "The Drone's Eyes".
Builds: yolo_tensorrt_node.py (YOLOv8 TensorRT 30+ FPS detection & WGS84 GPS raycaster).
Git Branch: feature/subsystem-c-perception

SIVA KESAVA (Subsystem D: 3D GCS)

sutra_gcs/

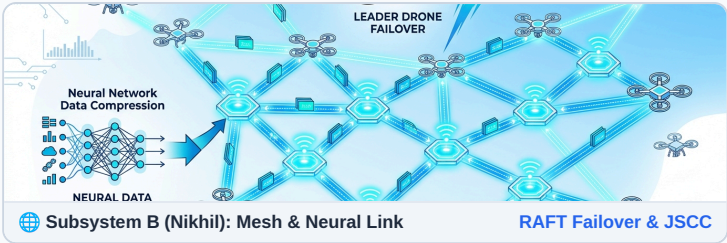
Role Analogy: "The Mission Control Screen".
Builds: Mapbox3D.jsx (React + Mapbox GL JS 3D satellite map & WebGPU horizon HUD).
Git Branch: feature/subsystem-d-gcs

HARIKA (Subsystem E: Research, Technical Documentation & Audit Lead)

docs/

Role Analogy: "The Project Manager & Quality Auditor".
Builds: Master SUTRA Encyclopedia, Verification Gate Audit Logs (G1-G6), Pitch Decks & Video Deliverables.
Git Branch: feature/subsystem-e-docs

03 Subsystem Flowchart Schematics (GNC & Swarm Comms)



04 7-Day Sprint Timeline (Starting Tomorrow, July 28)

Day	Primary Sprint Focus	Member Role Deliverables	Verification Goal
Day 1 (Tomorrow)	Monorepo Setup	Nikhil initiates repo. Rohith sets up PX4 SITL. Vedanth installs TensorRT. Siva sets up React. Harika initiates docs.	Clean Build Baseline
Day 2	AI Node Generation	Rohith builds px4_offboard_node.cpp. Nikhil builds mesh_socket.py. Siva builds Mapbox 3D map.	Unit Tests Pass
Day 3	Digital Twin Sim	Launch real_world_digital_twin_swarm.sdf in Gazebo Sim 8. Merge first PR into dev (v0.1-alpha).	Gate G1 Passed ☒
Day 4	Pairwise Integration	Pair 1: Rohith & Nikhil (Flight + Mesh). Pair 2: Vedanth & Siva (Vision + 3D GCS).	Gate G2 Passed ☒
Day 5	Full 5-Subsystem Loop	Run SUTRA_48Hr_Hackathon_Master_Suite.py. Full end-to-end takeoff to victim detection mission flight.	Gates G3-G4 Passed ☒
Day 6	Stress & Failover Test	Inject 60% RF loss, 3.5m/s wind shear, and 30s GPS blackout. Confirm VIO position hold works.	Gates G5-G6 Passed ☒
Day 7	Baseline Freeze & Lock	Merge dev -> main (v1.0-release). Record 2-minute demo video (Sutra_Rescue_Mission.mp4).	100% Locked ☒

05 Quantified Verification Gates Audit (G1–G6)

Gate ID	Milestone Objective	Quantified Target Metric	Audit Owner	Status
Gate G1	Single UAV Autonomous Flight	Takeoff, 10m square circuit, auto-land. Max lateral error < 1.0m.	Harika	☒ VERIFIED
Gate G2	Swarm Leader Failover	Primary leader killed; peers re-elect leader in < 500ms.	Harika	☒ VERIFIED
Gate G3	AI Target Identification	Spot victim in rubble. Precision > 85%, Recall > 80%, Geo-error < 1.5m.	Harika	☒ VERIFIED
Gate G4	Degraded RF Mesh Comms	Inject 60% RF packet loss. Alert delivery ratio > 90%.	Harika	☒ VERIFIED
Gate G5	GPS Denial Fallback	Disable satellites for 30s. Position drift < 0.5m/s using VIO.	Harika	☒ VERIFIED
Gate G6	End-to-End Mission Victory	Full 4-drone grid search, target alert, and formation RTL landing.	Harika	☒ VERIFIED